Mirco De Zorzi

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Skills

- **Languages**: Golang, Java, C#, Rust, Python, C++, C, Javascript, Lua
- **Databases**: Kafka, Prometheus, Loki, ElasticSearch, Redis, Cassandra, Postgres
- **Tools**: AWS, Kubernetes, Helm, Serverless, Ansible, Docker, gRPC, V8, Git, Linux
- **Skills**: Distributed Systems, Data Driven Architecture, Web API Reverse Engineering
- **Natural Languages**: Italian (native), English (proficient), Slovakian (second language)

Talks

- **KubeCon 2023**: Kubernetes Defensive Monitoring with Prometheus

Experience

Microsoft Azure — October 2024 — Present
Software Engineering L62 — *Prague, Czech Republic*

Developed and maintained Holmes, a large-scale distributed orchestration platform within Azure that optimizes resource utilization and ensures high availability through Live Migration and Service Healing.

- Designed and led the development of HolmesV2, a comprehensive overhaul of the existing architecture, focusing on a new query engine and a complete refactoring of candidate selection computation, driven by a deep understanding of the existing system's limitations and a focus on addressing key performance, scalability, and maintainability challenges.
- Led instrumentation and observability initiatives to improve the visibility of the system and reducing mean-time-to-detect for live-site incidents from hours to minutes, alongside easily pinpointing performance hotspots.
- Achieved a 90% reduction in database reads by identifying performance hotspots and developing a custom caching library.
- Optimized the component responsible for selecting candidates for action, reducing execution times through targeted refactoring and minimizing memory allocations, from peaks of 64GB/minute to 8GB/minute.
- Reduced memory usage by 50% by optimizing Kafka message ingestion, processing, and retention.
- Optimized the ingestion pipeline, reducing p95 ingestion delays from 13 seconds to 500ms.
- Optimized the CI/CD pipeline, reducing deployment times from over 2 hours to under 40 minutes.
- Introduced custom C# syntax analyzers to catch issues at compile time, significantly reducing runtime errors.
- Established and maintained a comprehensive best practices document, adopted by a team of over 20 people.

Sysdig — February 2021 — September 2024
Data Platform Engineer 2 — *Remote, Italy*

Developed and maintained Sysdig Monitor's real-time time-series database read path:

- Developed features for AST constant expression folding and push-down of complex expressions to the storage layer.
- Developed features to optimize a query's AST to prevent unnecessary computation and data retrieval
- Reduced query evaluation times by up to 50% by parallelizing AST sub-trees evaluation.
- Reduced network traffic between storage layer and query engine by up to 30% by optimizing the data transfer protocol.
- Reduced cpu and memory usage by 30% while increasing throughput by 30% by optimizing de-serialization logic.

Technological Systems by Moro — June 2019 - August 2019 & December 2020 - March 2021
Software Development & Penetration Testing & Reverse Engineering — *Azzano Decimo, Italy*

- Developed a monitoring framework based on a serverless architecture hosted on AWS.
- *Simultaneous localization and mapping* and *automated guided vehicles* in C++ & R.O.S.
- Carried out a network vulnerability assessment and penetration test.
- Reverse Engineered stripped firmware for an STM32 microprocessor.

Projects

- **bbc**: A port of the *microsoft/bond* compiler to ANTL4 and C#, with wire format breaking change detection.
- **Espresso**: A port of *ben-manes/caffeine* to C#.
- **FlagOps**: Fully fledged, distributed and highly-available infrastructure to facilitate Attack/Defence CTF competitions.
  - Complete CI/CD automation with Github Actions and deployment through AWS ECR, Kubernetes and Helm charts.
  - Cluster-aware service to deploy and orchestrate swarms of version controlled short-living containers;
  - A L7 reverse proxy and host based intrusion detection system written in Go leveraging network traffic analysis, system monitoring, and kernel level tracing to detect and block potentially malicious connections.
- **sentry**: High performance reverse proxy and load balancer written in C++ with support for Lua scripting.
- **hort**: Data mining and indexing framework written in modern C++.
- **pka2xml**: Packet Tracer utility tools obtained through reverse engineering.

Additional Experience & Achievements

- Participated at Hack-A-Sat 4 & DEFCON 31 *Capture The Flag* competitions with *mhackeroni* and placed respectively 1st and 10th place.
- CyberChallenge 2022 & 2023 Network Security and Hardening mentor at Ca'Foscari.
- Regularly participate and organize cyber-security related meeting at Ca'Foscari.
- Qualified at a national programming challenge organized by the *National Interuniversity Consortium for Informatics* at the University of Pisa and subsequently attended a 3 months long cybersecurity course at the University of Padua with topics such as: reverse engineering, system security, web security, and cryptography. At the end of this course participated at a jeopardy CTF hosted by the organizers of the course.
- Attended a week-long *advanced algorithms* course at San Servolo International College for students qualified at the 2019 National Olympiad of Informatics. The topics ranged from dynamic programming to greedy algorithms, and graphs.
- Won the M.I.U.R. prize at the *National Robotic Olympics* of 2019 with a remotely operated drone for environmental data collection, processing, and analysis. Presented the prototype at the Italian Ministry of Education during the National Day for the Environment.
- Participated at 3 national Rubik's cube competitions in Milan, Verona, and Bratislava (SK) with a personal best of 18.67 seconds.

Education

Ca'Foscari — Venice, IT
Bachelor's in Computer Science — *2021 — present*

I.T.I.S J.F.Kennedy — Pordenone, IT
Diploma in Information Technologies — *2016 — 2021*